THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 7: **Arithmetic Progressions, Coordinate Geometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 1.1	$1.7 - 0.6$
2	**(a)** 99	$4 + 19 \times 5$
3	**(c)** 13th	$2 + 4(n-1) = 50$
4	**(b)** Incorrect — $d = 0$	$3 - 3 = 0$
5	**(d)** 400	$n^2 = 400$
6	**(b)** $5\sqrt{2}$	$\sqrt{25+25}$
7	**(d)** $(1, 1)$	$\left(\frac{2}{2}, \frac{2}{2}\right)$
8	**(a)** $(2, 3)$	$\left(\frac{6}{3}, \frac{9}{3}\right)$
9	**(a)** Incorrect — they are collinear (on $y = x$)	$\sqrt{2} + \sqrt{2} = 2\sqrt{2}$
10	**(a)** Student A is correct	Distance from x-axis $= \lvert y \rvert$; 5 is distance from origin

SECTION — B (Very Short Answer) — Answers 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a_n = 1 + 3(n-1) = 3n - 2$; $a_{25} = 73$	1+1
12	$S_{12} = 6(10 + 44) = 324$	1+1
13	$9 + (k-3)^2 = 25 \Rightarrow k - 3 = \pm 4 \Rightarrow k = 7$ or -1	1+1
14	$\left(\frac{7+2}{3}, \frac{9+6}{3}\right) = (3, 5)$	1+1

SECTION — C (Short Answer) — Answers 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$4d = 16 \Rightarrow d = 4$; $a + 28 = 31 \Rightarrow a = 3 \Rightarrow 3, 7, 11, 15, \ldots$	1+1+1
16	$\frac{5k-3}{k+1} = -1 \Rightarrow k = \frac{1}{3} \Rightarrow 1 : 3$ (check y: $\frac{3+1}{4/3} = 3$ ✓)	1+1+1
17	$105, \ldots, 994$: $\frac{994-105}{7} + 1 = 128$	1+1+1
	OR: $306, \ldots, 693$: $n = 44$; $S = 22(306 + 693) = 21978$	1+1+1
18	All sides $= \sqrt{17}$; diagonals $AC = BD = \sqrt{34} \Rightarrow$ square	2+1
	OR: $(x+2)^2 + 25 = (x-2)^2 + 9 \Rightarrow 8x = -16 \Rightarrow x = -2 \Rightarrow (-2, 0)$	1+1+1

SECTION — D (Long Answer) — Answers 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $50 + 9 \times 20 = ₹230$ (ii) $5(50 + 230) = ₹1400$ (iii) $S_n = 10n^2 + 40n = 2850 \Rightarrow n^2 + 4n - 285 = 0 \Rightarrow n = 15$ weeks	1+2+2
	OR: $a = 8$, $S_2 = 26 \Rightarrow a_2 = 18$, $d = 10$; $a_{20} = 198$; $8 + 10(n-1) = 108 \Rightarrow n = 11$	2+1+2
20	(i) 4, 3 (ii) both 5 (iii) rectangle (opposite sides equal, diagonals equal, adjacent sides unequal) (iv) $(1, 0.5)$	1+2+1+1
	OR: $\frac{24a+3a+1}{4} = 9a - 2 \Rightarrow a = 1$; $\frac{15-3}{4} = 3 = -b \Rightarrow b = -3$; A(4, −3), B(8, 5): $AB = 4\sqrt{5}$	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5 & 7: **Arithmetic Progressions, Coordinate Geometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** -4	$d = -3$
2	**(d)** $3, 5, 7$	$n = 1, 2, 3$
3	**(b)** 31	$\frac{90}{3} + 1$
4	**(c)** Incorrect — $a + c = 2b$	$b - a = c - b$
5	**(a)** Student A is correct	15×16
6	**(b)** 5	$\sqrt{16 + 9}$
7	**(b)** fourth quadrant	$x > 0,\ y < 0$
8	**(a)** $(4, 3)$	$a + 2 = 6,\ 4 + 2b = 10$
9	**(c)** Incorrect — it is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Distance formula
10	**(b)** Student A is correct	$\frac{4k-2}{k+1} = 0 \Rightarrow k = \frac{1}{2}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a_{31} = 305$; $5 + 10(n - 1) = 435 \Rightarrow n = 44$	1+1
12	$6(1 + 2 + \cdots + 40) = 6 \times 820 = 4920$	1+1
13	(i) 13 (ii) 5	1+1
14	$\frac{3+k}{2} = 4 \Rightarrow k = 5$	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$d = -3$, $n = 76$; $S = 38(-235) = -8930$	1+1+1
16	$\frac{-5k+3}{k+1} = 0 \Rightarrow k = \frac{3}{5}$ ⇒ 3 : 5; point $\left(\frac{-6+30}{8}, 0\right) = (3, 0)$	1+1+1
17	$a + 8d = 0$; $a_{29} = a + 28d = 20d$, $a_{19} = a + 18d = 10d$ ⇒ $a_{29} = 2a_{19}$	1+1+1
	OR: From the end: $201 - 11 \times 2 = 179$	1+1+1
18	$AB = 4$, $AC = BC = 2\sqrt{2}$; $AC^2 + BC^2 = 16 = AB^2$ ⇒ right isosceles triangle	2+1
	OR: $16 + (3 - y)^2 = 25 + (2 - y)^2 \Rightarrow y = -2$ ⇒ $(0, -2)$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{n}{2}[40 + 15(n - 1)] = 3250 \Rightarrow 3n^2 + 5n - 1300 = 0$ (ii) $D = 125^2$ ⇒ $n = 20$ (iii) $20 + 19 \times 15 = ₹305$	2+2+1
	OR: (i) $n = 38$, $S = 19 \times 367 = 6973$ (ii) 19th and 20th terms: 179 and 188	3+2
20	(i) 10 (ii) $\left(\frac{17}{5}, \frac{26}{5}\right) = (3.4, 5.2)$ (iii) $F = (13, 18)$ (iv) 20	1+2+1+1
	OR: $AB^2 = BC^2 = 106$, $AC^2 = 212 = AB^2 + BC^2$ (3). Mid-point of AC $= (5, 3)$: distances to A, B, C all $\sqrt{53}$ (2)	3+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 7: **Arithmetic Progressions, Coordinate Geometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $-2, -2, -2, \ldots$	Constant terms
2	**(b)** 59	$3 + 14 \times 4$
3	**(a)** 5050	$\frac{100 \times 101}{2}$
4	**(a)** Incorrect — the differences 3, 5, 7 are not equal	Not a constant difference
5	**(a)** Student A is correct	Coefficient of n
6	**(b)** $a\sqrt{2}$	$\sqrt{a^2 + a^2}$
7	**(b)** $(4, -3)$	Halve each coordinate
8	**(b)** $(0, 3)$	Opposite to $(4, 0)$
9	**(a)** Correct — both distances are 5	$\sqrt{9 + 16} = \sqrt{16 + 9}$
10	**(a)** Student A is correct	Mid-point

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$31 - 3(n-1) = 0 \Rightarrow n = \frac{34}{3}$, not natural ⇒ No	1+1
12	$a_1 = 4,\ a_{25} = -68;\ S = \frac{25}{2}(4 - 68) = -800$	1+1
13	$(x-7)^2 + 36 = (x-3)^2 + 16 \Rightarrow 8x = 60 \Rightarrow x = \frac{15}{2}$ ⇒ $\left(\frac{15}{2}, 0\right)$	1+1
14	$\left(\frac{18-3}{5}, \frac{16+9}{5}\right) = (3, 5)$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$n^2 - 37n + 342 = 0 \Rightarrow n = 18$ or 19; the 19th term is 0, so both sums are 513	1+1+1
16	Equating distances: $k = -2$, then $h = 3$ ⇒ centre $(3, -2)$	1+1+1
17	$a - 3d, a - d, a + d, a + 3d$: $a = 5$; $100 + 20d^2 = 120 \Rightarrow d = \pm 1$ ⇒ 2, 4, 6, 8	1+1+1
	OR: $a_{n+1} - a_n = -5$ (constant) ⇒ AP; $a_1 = 4$, $S_{15} = \frac{15}{2}(8 - 70) = -465$	1+2
18	$-12x + 8y = 0 \Rightarrow 3x = 2y$	1+1+1
	OR: Mid-points: BC (1, 2), CA (0, 1), AB (1, 0). Medians: $\sqrt{10},\ 2,\ \sqrt{10}$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $20 + 58 = 78$ (ii) $15(20 + 78) = 1470$ (iii) $20 + 2(n-1) = 50 \Rightarrow n = 16$	2+2+1
	OR: $(p-q)[2a + (p+q-1)d] = 0 \Rightarrow 2a + (p+q-1)d = 0 \Rightarrow S_{p+q} = 0$ (3). $S_{17} = \frac{17}{2}[48 + 16(-3)] = 0$ ✓ (2)	3+2
20	(i) 10 (ii) $(4, 5)$ (iii) $(3, 1)$ and $(5, 1)$ (iv) both distances 5 ✓	1+1+2+1
	OR: (i) $OA = 2$: $(x-2)^2 + 4 = 4 \Rightarrow x = 2$ (ii) $(0, 3)$ (iii) distance from O $= 2$ ⇒ yes	2+2+1

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ANSWER KEY & MARKING SCHEME SET D

Chapters 5 & 7: **Arithmetic Progressions, Coordinate Geometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $10, 20, 30, 40$	Add 10 each time
2	**(a)** 5	$\frac{1}{2} + 9 \times \frac{1}{2}$
3	**(b)** 21	$24 - 3$
4	**(b)** Incorrect — it is $\frac{n(n+1)}{2}$	Sum of an AP
5	**(b)** Student A is correct	$12, \ldots, 248$: $\frac{236}{4} + 1 = 60$
6	**(a)** 5	$\sqrt{9+16}$
7	**(a)** $(1, 2)$	$\left(\frac{2}{2}, \frac{4}{2}\right)$
8	**(b)** $(6, 0)$	$(x-2)^2 + 9 = 25 \Rightarrow x = 6$ or -2
9	**(a)** Incorrect — the abscissa is 0	Points $(0, y)$
10	**(b)** Student A is correct	$\frac{3-3}{2} = 0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	100 terms ⇒ $100^2 = 10000$	1+1
12	$4d = 12 \Rightarrow d = 3$, $a = 1$	1+1
13	Each side squared $= 8$ (e.g. $(1+\sqrt{3})^2 + (\sqrt{3}-1)^2 = 8$) ⇒ equilateral	2
14	$B = (10-2,\ -2-3) = (8, -5)$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Middle angle $= 60^\circ$ (sum 180°) ⇒ $d = 20^\circ$ ⇒ $40^\circ, 60^\circ, 80^\circ$	1+1+1
16	$\frac{3k-6}{k+1} = -4 \Rightarrow k = \frac{2}{7}$ ⇒ 2 : 7 (check $y = 6$ ✓)	1+1+1
17	$3a_{19} = 225 \Rightarrow a + 18d = 75$; $3a_{36} = 429 \Rightarrow a + 35d = 143$ ⇒ $d = 4,\ a = 3$ ⇒ $3, 7, 11, \ldots$	1+1+1
	OR: $11, 13, \ldots, 99$: 45 terms; $S = \frac{45}{2}(110) = 2475$	1+1+1
18	Mid-points of AC and BD coincide: $a = 1,\ b = 3$	1+1+1
	OR: $P = (3, -2)$ ⇒ $6 + 2 + k = 0 \Rightarrow k = -8$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $30 - 2(n-1) = 2 \Rightarrow n = 15$ (ii) $\frac{15}{2}(32) = 240$ (iii) $30 - 18 = 12$	2+2+1
	OR: $a + 4d = 3$; $(3-2d)(3+2d) = 8 \Rightarrow d = \pm\frac{1}{2}$. $d = \frac{1}{2}, a = 1$: $S_{16} = 76$; $d = -\frac{1}{2}, a = 5$: $S_{16} = 20$	3+2
20	(i) 10 (ii) $\left(\frac{28}{5}, \frac{39}{5}\right)$ (iii) $(x-2)^2 + 9 = (x-8)^2 + 121 \Rightarrow x = \frac{43}{3}$ ⇒ $\left(\frac{43}{3}, 0\right)$	1+2+2
	OR: $AB = CD = \sqrt{17}$, $BC = DA = \sqrt{2}$ ⇒ parallelogram (3). $AC = \sqrt{13} \neq BD = 5$ ⇒ not a rectangle (2)	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.